# Manual

## PDV Archiving

## PDV-ARC 8.1

Version 1.0.23372

Last changed on: 23.09.2020

## Copyright

# Contents

# 1 Process data management - archiving

## Overview

## Purpose

PDV archiving is divided into two areas:

- Archiving the raw data, also referred to as mass data
- Archiving other PDV data

Mass data archiving makes use of a stand-alone archiving process, the other data are archived in the manner they are generally handled in HYDRA data management.

## Features

Configuration functions and administrative operations for controlling the archiving of process values.

- Management of archive data across adjustable periods in the HYDRA database.
- Archiving of raw measured value data by exporting them into the archive system based on predefined amounts
- Automatic import of data from the archive system when a report is created for an exported period
- Direct access to the archived process data using the functions in the HYDRA console
- Two-level archiving of order/article information associated with data collection, as well as of events, process disturbances, target value changes and other transaction data. Medium term area is located within the database and for that reason is immediately accessible. The exported long-term information is temporarily retrieved by the reload manager.

# 2 Data Management

## Overview

| | |
|---|---|
| HYDRA menu | System administration → Archiving → Data management |
| FEDRA menu | System administration → Archiving → Data management |
| Transaction code | arccfg |
| Function authorization | arccfg.* |

## Purpose

You use this application to view or change the centrally managed archiving settings of recorded data.

## Integration

Use this application to configure the settings for archiving / data management for all components/functions.

## Field descriptions

**Product**

Enter the system module for which you want to define a rule.

**Object**

Use the *Object* to define the data you want to retain.

**Retention period**

Use the fields for the *Retention period* to define how long data should be available until data is archived.

**Unit**

Specify the retention period in days, months or years.

**Last retention date**

This field is computed and specifies the last day when data is still available in this object.

**Action**

You can choose from three options to specify the processing for an object:

| | |
|---|---|
| D = delete | Object is deleted |
| M = move (archive) | Object is transferred to the next area |
| X = export | Object is unloaded (XUNLOAD format) and then deleted |

**Target object**

Currently, the target object automatically results from the detail configuration.

**Condition**

The characters in this field are added as an additional condition to the database command that controls the action. This condition is linked to the other conditions of the original command using AND and is set in parentheses.

**Last run (date/time)**

Indicates the point in time when this rule was applied last.

**License**

Indicates which license is required for archiving. You can enter several licenses (separated by a space). If none of these licenses is licensed in the system, the data is either deleted (action D) or unloaded into files (action M or X), depending on the relevant action.

**Path**

Optional path to generate the file export (unload). The archiver currently only supports local HYDRA server drives. If you do not enter a path, file exports are filed as follows:

**<HYDRADIR>/<SYSTEM>/custom/archive/<YYYY-MM-DD>/<PRODUCT>/**

HYDRADIR HYDRA directory
SYSTEM … system number
YYYY-MM-DD … archiving day (YYYY … year, MM … month, DD … day)
PRODUCT … product from archiving configuration

**Administration table**

Name of the administration table where archiving logs are stored.

**Administration duration (retention period of administration table)**

Use the fields for the *retention period* to specify how long the logs should be available in the administration table before being deleted.

**Unit**

Specify the retention period in days, months or years.

**Archiving step**

Specifies if this archiving process uses archiving function I or II. Archiving function I: The function moves data from the online data set to archive tables or deletes the data: setting M (medium-term archive). Archiving function II: The function moves data from the archive table to the file export or deletes the data: setting L (long-term archive).

**Configuration**

Indicates whether or not the configuration is active. Possible values: Y/N

**Archiving type**

Identifier for time or object-related archiving. Supported modes: O = Object-related archiving (i.e. data is archived for each object individually). Z = Time-related archiving (i.e. data is archived without any object reference).

Note: Time and object-related archiving differ from each other significantly in the archiving performance (runtime). Object-related archiving of mass data is not recommended.

**Priority**

Integral value greater than 0. Indicates the processing sequence if several objects are defined for a product group. Processing starts with the lowest value.

**Master table**

Table including the data to be archived. The extensions entered in the *Condition* field refer to this table.

**Date column**

Date column in the master table. The system uses this date column to evaluate the retention period for the data to be archived.

**Key 1**

Unique key column in the master table; this key identifies the data to be archived. You can define up to 5 key columns for object-related archiving. Time-related archiving only supports one primary key in *Key 1.*

**Keys 2 – 5**

Additional optional key columns for object-related archiving.

**Comment**

Use the comment line to describe the archiving configuration.

If you copy the archiving configuration, the system currently only copies the data management configuration, but no defined data records from the object details. Currently, you must use the respective dialog to copy the data records of the object details to the new configuration.

# 3 Functions/configurations specific to product groups

## 3.1 CAQ

For information on the data management configurations of CAQ 8.1, refer to the document MBL_Archiving_CAQ.pdf.

## 3.2 BDE

For information on the data management configurations of BDE 8.1/BDE 8.2, refer to the document MBL_Archiving_BDE.pdf.

## 3.3 MDE

For information on the data management configurations of MDE 8.1/MDE 8.2, refer to the document MBL_Archiving_MDE.pdf.

## 3.4 WRM

For information on the data management configurations of WRM 8.1/WRM 8.2, refer to the document MBL_Archiving_WRM.pdf.

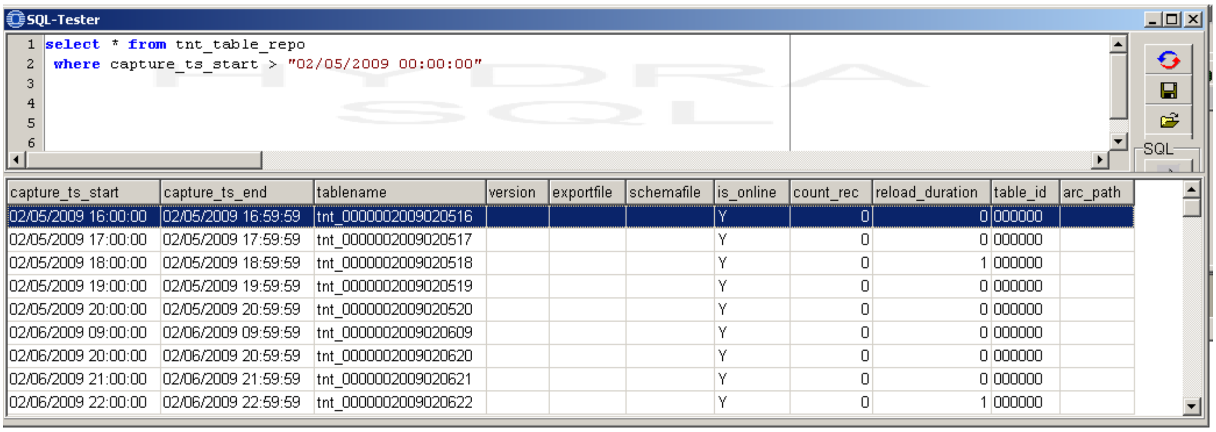
# 3.5 PDV

## 3.5.1 Overview

The standard archiving function of HYDRA-PDV 7.2 can archive mass data tables. The system writes table data into files that are stored using a defined path.

Mass data tables (also called TNT tables) include a time stamp in their name ID that is used during archiving.

The table "tnt_table_repo" provides information on the archiving status of the single TNT tables, e.g. information on the location and the name of the data file, or on an eventual re-import of the table that might require a new export.




```
select * from tnt_table_repo
 where capture_ts_start > "02/05/2009 00:00:00"
```

| capture_ts_start | capture_ts_end | tablename | version | exportfile | schemafile | is_online | count_rec | reload_duration | table_id | arc_path |
|---|---|---|---|---|---|---|---|---|---|---|
| 02/05/2009 16:00:00 | 02/05/2009 16:59:59 | tnt_0000002009020516 | | | | Y | 0 | 0 | 000000 | |
| 02/05/2009 17:00:00 | 02/05/2009 17:59:59 | tnt_0000002009020517 | | | | Y | 0 | 0 | 000000 | |
| 02/05/2009 18:00:00 | 02/05/2009 18:59:59 | tnt_0000002009020518 | | | | Y | 0 | 1 | 000000 | |
| 02/05/2009 19:00:00 | 02/05/2009 19:59:59 | tnt_0000002009020519 | | | | Y | 0 | 0 | 000000 | |
| 02/05/2009 20:00:00 | 02/05/2009 20:59:59 | tnt_0000002009020520 | | | | Y | 0 | 0 | 000000 | |
| 02/06/2009 09:00:00 | 02/06/2009 09:59:59 | tnt_0000002009020609 | | | | Y | 0 | 0 | 000000 | |
| 02/06/2009 20:00:00 | 02/06/2009 20:59:59 | tnt_0000002009020620 | | | | Y | 0 | 0 | 000000 | |
| 02/06/2009 21:00:00 | 02/06/2009 21:59:59 | tnt_0000002009020621 | | | | Y | 0 | 0 | 000000 | |
| 02/06/2009 22:00:00 | 02/06/2009 22:59:59 | tnt_0000002009020622 | | | | Y | 0 | 1 | 000000 | |

**Patch:**

The standard HYDRA-PDV 7.2 installation provides the option to archive mass data, thus TNT tables.

- DBPATCH PDV_72 (includes the “pdv_setup“ table)
- DBPATCH TNT_72 (includes the “tnt_table_repo“ table)

**License:**

| Lizenz | Produktbezeichnung | Kategorie | Produkt | Anzahl | Gültig bis | Release |
|---|---|---|---|---|---|---|
| | PDV | HYDRA | PDV-ARC | 1 | | 7.2 |

## 3.5.2 Configuration

Mass data is archived via two components: export and import.

The export is performed at cyclic intervals and is started by the system scheduler. The export unloads “due” tables from the database into data files on hard disk.

The import (or reload) function is a kind of library. If required, you use this function to reload data from the data files back to the database tables.

In general, the export function transfers the files one-to-one from the database into the data files. You can also configure a one-to-many relationship, i.e. you can distribute a table among several files. Vice versa, a file must belong to exactly one table (one table can have several files, but a file always refers to exactly one table).

| | |
|---|---|
| Program: | hp_mexp.exe / out |
| Installation: | The program is entered in the system Scheduler and started cyclically. |
| | Console menu: File – System administration – Scheduler |
| | The corresponding time, when the program is to be started can be defined in the “fix” tab. |

**Important:** You should not start the export program at little intervals in the Scheduler, because it is an archiving program. Depending on the data volume, a daily (each night) or weekly rhythm is enough.

You must also define a parameter that specifies when the data is old enough to be archived. Here, the parameter refers to the number of tables that may be online. If this number is exceeded, the oldest TNT tables are archived via export. The parameter is included in the “pdv_setup” table. You can change the parameter in the basic settings of the console, in tab “PDV”.

As part of the implementation process, it has to be checked if the archiving path with ID “PDVARC“ exists in the “arc_path” column of the “pdv_setup” table (and, if required, if the transport path has been defined with the ID “PDVTRANS” in the “trans_path“ column).

You can then use the table "hy_path" to find the respective directories.

You may not define a host if a relative URL path is defined for archiving, PDVARC, (depending on the file hp_mexp.scr).

As part of a customization, you can also archive further data ( e.g. events). To this end, you must store the respective entries in the table "hyd_datamanagement". The process is identical to the one of the general archiving.

## 3.6 MPL / TRT

For information on the data management configurations of MPL or TRT in versions 8.1 or 8.2, refer to the document MBL_Archiving_MPL.pdf.

## 3.7 HLS

For information on the data management configurations of HLS 8.1, refer to the document MBL_Archiving_HLS.pdf.

## 3.8 PZE / PZW

For information on the data management configurations of PZE or PZW in versions 8.1 or 8.2, refer to the document MBL_Archiving_PZW.pdf.

## 3.9 PEP

For information on the data management configurations of the Personnel Scheduling PEP in versions 8.1 or 8.2, refer to the document MBL_Archiving_PEP.pdf.

## 3.10 LLE

For information on the data management configurations of the Incentive Wage LLE 8.1, refer to the document MBL_Archiving_LLE.pdf.

## 3.11 ZKS

For information on the data management configurations of the Access Control ZKS in versions 8.1 or 8.2, refer to the document MBL_Archiving_ZKS.pdf.

## 3.12 ESK

For information on the data management configurations of the Escalation Management in version 3.0, refer to the document MBL_ESK_Archiving.pdf.

## 3.13 ETD

For information on the data management configurations of the Label Design in version 3.0, refer to the document MBL_Archiving_ETD.pdf.

# 4 Reload Manager

## Summary

| | |
|---|---|
| HYDRA menu | System administration → Archiving → Reload manager |
| FEDRA menu | System administration → Archiving → Reload manager |
| Transaction code | arcrld |
| Function authorization | arcrld<br>arcrld.export<br>arcrcl.import |

## Purpose

The customer is responsible for backing up data in a data archive and for restoring data. When data is backed up it is transferred from the long-term data area to the archive data area and, as a result, it is stored in a separate file system. The customer can start the backup process using the Reload Manager.

When data, which is filed in the archive data area, is restored, it is transferred back (copied) to the long-term data area. The customer has to do this and bears the responsibility.

To be able to evalaute restored data using standard evaluations/reports, the data has to be transferred back (copied) to the reload data area.

For this reason, the Reload Manager enables the following functions:

1. Moving of exported data into the customer archive (the customer archive path has to be defined within the HYDRA path settings).
2. Loading of exported data into the reload area for evaluations

## Integration

The Reload Manager is a central function that is used by many components or functions.

## Selection Criteria

The application provides the following selection criteria:

**Module**

Reference to the product group which archived data belongs to.

**Object type**

Object type of the archived data.

The selection options “module” and “object type” are mandatory fields.

# Toolbar

**Export**

The “EXPORT” button allows for the entries selected in the Reload Manager to be exported to the customer directory (all data records can be selected using the context menu of the right mouse button). Once the function has been started, the input dialog that opens requires the customer archive path to be entered.

**Import**

The “IMPORT” button allows for the selected entries to be loaded into the reload area, in order for the data to be again available for HYDRA evaluations/reports. Once the function has been started, the data loading mode and an optional path are requested. By indicating the path, it is possible to specify another storage location for the files, provided that the archived files have not been moved to the customer-specific archive using the Reload Manager.

There are three different ways to deal with reload data after they have been reloaded to prevent the reload data set from increasing excessively (→ no slow evaluations/reports):

The following modes are distinguished:

- Cyclic
  Data is loaded to the reload area in the “cyclic” mode. Loaded data is automatically removed from the corresponding reload tables, once the retention period specified within archive settings has expired.
  Please compare the configuration entry HYD / RELOADMANAGER of the data management function. When demo settings are used, data is removed from the reload area after 14 days.

  When data is imported, the HYD_REL_MANAGEMENT.DELETE_DATE field is calculated subject to the configuration and entered in the reload management table.

- User-specific
  Data is loaded to the reload area in the “user-specific” mode. The loaded data is automatically removed from the corresponding reload area, once the time specified for the user within the user-specific settings has expired. Exception: In case several users loaded identical data into the reload area, the corresponding retention period is automatically set to the maximum date.
  **When data is imported, the HYD_REL_MANAGEMENT.DELETE_DATE field is computed subject to the configuration and entered in the reload management table.**

- Manual
  Data is loaded in the “manual” mode to the reload area. The customer is responsible for deleting data from corresponding reload tables. However, identical data cannot be loaded several times in the “manual” mode (data can be loaded in the “user-specific” mode though).

In order for data to be transferred to the reload data area, the HYDRA server must be able to access these files. Otherwise, loading is cancelled with an error message.